# Hazard Analysis
# Flexin

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Table 1: Revision History

| **Date** | **Developer(s)** | **Change** |
|---|---|---|
| Oct 7th, 2026 | Noah Goldschmied | Wrote Sections 1-2 |
| Oct 8th, 2026 | Noah Goldschmied | Wrote Sections 3-4 |
| Oct 9th, 2026 | Noah Goldschmied | Wrote Sections 5, 6, Appendices |
| Oct 18th, 2026 | Noah Goldschmied | Wrote roadmap & final changes |

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# 1 Introduction

This document aims to illustrate the hazards that may occur during the development and deployment of Flexin'. It defines hazards for this project, the purpose of analyzing them, and how the team will mitigate them to ensure safe use of Flexin'.

## 1.1 Flexin' Problem Statement

Personal finance and fitness follow the same pattern: the payoff for effort put in today shows up years later. Someone who invests $100 a month from age 25, assuming a 7% annual return, contributes $48,000 over 40 years and ends up with roughly $262,000. Waiting until 35 to start the same plan leaves them with about $122,000. The same logic applies to exercise. A single walk or workout changes very little, but the habit, kept up for years, is what separates a healthy 40-year-old from an unhealthy one. In both cases the early actions matter most, and in both cases they are the easiest to skip, because the reward is too far away to feel real.

The concepts behind this are not widely understood, and young people understand them the least. A 2012 survey of 6,911 Canadians asked three basic questions covering compound interest, inflation, and risk diversification [1]. Only 42% of respondents answered all three correctly. Among respondents under 35, that number fell to about 30%, the lowest of any age group, and 44% of them answered "don't know" to at least one question. Only about half of respondents under 35 knew that buying a single company's stock is usually riskier than buying a stock mutual fund. The same study found that respondents who answered all three questions correctly were 10 percentage points more likely to have retirement savings, which suggests that this gap in knowledge carries through to real financial outcomes.

Tools for practising each skill already exist, but separately. Stock market simulators let people trade with fake money, and fitness apps reward steps with points and badges. A market simulator only helps someone who is already motivated to learn about investing, which excludes most of the young adults the survey shows are furthest behind. Fitness rewards, meanwhile, rarely lead to anything beyond the app itself. No product currently connects the two so that a habit a user is already building (or wants to build) becomes their way into learning how money grows.

## 1.2 Hazard Analysis Introduction

A hazard as it pertains to Flexin' is a condition in the software that can create an unwanted effect of event on Flexin's users, or the general environment. In software, these hazards stem from poor design decisions or requirements. These hazards must be addressed prior to and during development in order to limit any unwanted impact on users.

As described above, Flexin' focuses on developing healthy habits for investing and fitness and teach users the effect of compounding. However, to accomplish this goal there are many hazards that may hurt users physically, psychologically, or financially. These hazards may come from poor software implementation, incompatible data, or faulty integration with outside sources. The hazard analysis document below aims to identify all hazards and how the team plans to address them.

## 2 Scope and Purpose of Hazard Analysis

The hazard analysis document outlines the hazards that may occur while using Flexin'. The high level hazards in scope of Flexin' include:

- **Physical/Psychological Harm**: Includes injuries that may occur during workouts that users intend to track. Also includes any impacts on mental health caused by the platform, such as addiction (similar to Instagram or TikTok) or feelings of embarrassment/shame for missing goals.
- **Financial Loss**: It is imperative to note that any interactions with financial markets on Flexin' are not indicative of real market results. Users may take strategies learned on the platform and incur serious financial loss.
- **Data Mismanagement**: Poor management of user data from database or cybersecurity perspectives may result in data leaks or loss of data. In addition, users improperly providing data can cause over/under rewarding of virtual currency, and can therefore lead to poor learning.

The purpose of the hazard analysis is to identify and mitigate risks that could affect the reliability, correctness and safety of the Flexin' app. Conduction of this analysis will allow the team to anticipate failures early, and design around the hazards in a way that prioritizes the safety of the users. The findings from this process will inform safety and security requirements, as well as having an impact on compliance and maintainability requirements. In addition, this work will contribute to the creation of a fun, safe, and intuitive platform for users to learn about compounding and the importance of fitness and financial literacy.

## 3 System Boundaries and Components

Flexin' will consist of many key components, all of which will come together to create the product. These components may not be separate entities, but they are distinct enough to present their own hazards.

- **Component 1: Account Creation & Authentication**
  - **Purpose:** Manages account creation and authentication for users
  - **Key Functions:**
    - Allow users to create accounts for Flexin'
    - Populate the database with the user account to store the created account
    - Store authentication key to ensure that only the account owner can log into their Flexin' account
- **Component 2: User Information**
  - **Purpose:** Manages user profile data such as name, height, weight, goals, workout data, etc.
  - **Key Functions:**

        * Read and write to the database with user information
        * Provide profile data for users
        * Allow users to set preferences within Flexin'

- **Component 3: Fitness Device Integrations**
    - **Purpose:** Allow Flexin' to connect with users' fitness devices
    - **Key Functions:**
        * Connect with users' devices
        * Take data from said devices when relevant
        * Ensure that connections with the devices are secure and reliable

- **Component 4: Exercise Data**
    - **Purpose:** Track and analyze users' exercise data
    - **Key Functions:**
        * Interact with the data collected from the fitness devices
        * Analyze exercise data against user goals
        * Read and write to the database with results and analytics

- **Component 5: Currency Attribution**
    - **Purpose:** Turn exercise results into virtual currency
    - **Key Functions:**
        * Use the analytics from exercise data
        * Compute the correct amount of currency to reward users
        * Read and write to the database with users' balances

- **Component 6: Market Simulation**
    - **Purpose:** The underlying behaviour of the market in Flexin'
    - **Key Functions:**
        * Teach users about the effects of compounding
        * Mimic proper stock market behaviour
        * Incorporate randomness through seeding

- **Component 7: Investment Platform**
    - **Purpose:** Allow users to invest in the simulated market with their currency
    - **Key Functions:**
        * Provide users with securities to invest in
        * Allow for buying and selling of said securities
        * Read and write to the database with investment information

- **Component 8: Social Management**

  - **Purpose:** Create a space for users to compete with their friends on Flexin'
  - **Key Functions:**
    * Manage group creation, validation and invites
    * Use the simulated markets randomness to create fair competition
    * Read and write to the database with group information

- **Component 9: Database**
  - **Purpose:** The single source of truth for all aspects of Flexin'
  - **Key Functions:**
    * ACID database that will ensure all data is correct
    * Accept reads and writes from components

- **Component 10: User Interface & Experience**
  - **Purpose:** Provide users a place to interact with all backend components of Flexin'
  - **Key Functions:**
    * Manage all user facing screens, navigation, and input
    * Ensure that components connect well to create a simple experience

# 4 Critical Assumptions

The below assumptions define the scope of the hazard analysis by stating conditions that are expected to hold. These conditions will allow the FMEA to go into depth about failure modes that Flexin' has control over.

- **Fitness Devices Properly Track Workouts**: Flexin' will assume that data provided by fitness devices is correct. Hazards that stem from false data originating from workout devices are out of scope for this document.

- **Users Understand Exercises They Attempt**: Flexin' assumes that users will not attempt workouts that they are either physically incapable of or do not understand proper form of. Physical hazards that stem from improper workouts are out of scope for this document.

- **Users Have Positive Intentions**: Flexin' assumes that users want to learn about compounding and build positive habits. If the user has negative intentions, Flexin' will not be able to provide value.

- **Third-Party Health Platforms Remain Available**: Flexin' assumes that the health-data APIs it depends on (e.g., Health Connect, manufacturer SDKs) remain supported and keep their documented behaviour. Hazards from these platforms being shut down or changing their terms of use are out of scope; temporary outages and malformed responses are still analyzed.

- **The User's Device Is Not Compromised**: Flexin' assumes that the device running the app has not been rooted, jailbroken, or infected with malware. Hazards from a compromised operating system intercepting or changing app data are out of scope.

- **Cloud Infrastructure Provides Its Advertised Guarantees**: Flexin' assumes that its hosting and database provider meets its stated durability and encryption-at-rest guarantees. Application-level failures such as bad writes, failed syncs, and misconfigured access rules are still analyzed.
- **Virtual Currency Has No Monetary Value**: Flexin' assumes that its virtual currency and simulated portfolios cannot be converted to or withdrawn as real money. Hazards from the virtual currency itself being lost or stolen are therefore out of scope; losses from users applying simulated results to real investments are still analyzed.
- **Users Provide Truthful Personal Information**: Flexin' assumes that the height, weight, fitness level, and health conditions users enter are accurate to the best of their knowledge.
- **Users Keep Their Credentials Private**: Flexin' assumes that users do not deliberately share their login credentials with others. Account takeover through stolen or leaked credentials is still analyzed.

# 5 Failure Mode and Effect Analysis

In this section, we will be analyzing the failure modes and effects of the system. Table 2 will break down the potential failure modes, their causes, effects, recommended actions, and the safety requirements that they are associated with.

Table 2: Failure Mode and Effect Analysis (FMEA)

| **Component** | **Failure Mode (Risk)** | **Effects of Failure(s)** | **Causes of Failure(s)** | **Recommended Action(s)** | **SR** | **Ref.** |
|---|---|---|---|---|---|---|
| Account Creation & Authentication | The system does not allow a user to create or sign into an account (Low) | • A new user cannot join Flexin'.<br>• A current user cannot sign into the app to view their progress. | • A technical failure, such as a database outage or loss of internet connection, prevents the request from completing.<br>• The user is locked out after too many failed login attempts.<br>• A bug in the sign-up or login flow rejects valid input. | • Database and cloud connectivity should be verified through CI before each release.<br>• Users should be able to reset their password through email.<br>• The app should show clear error messages explaining why sign-in failed and how to resolve it. | SR1, SR4 | H1.1 |
| Account Creation & Authentication | The system provides unauthorized access to a different user account (High) | • An attacker can view the account owner's private health, fitness, and portfolio data.<br>• An attacker can modify the account owner's goals, settings, and investments without their knowledge.<br>• The account owner may lose trust in Flexin' and stop using the platform. | • The user chooses a weak or reused password that is easily guessed.<br>• The application fails to sign users out, leaving sessions open on shared or lost devices.<br>• The user has not enabled multi-factor authentication to protect their login. | • The system should enforce a minimum password strength during account creation.<br>• The system should offer optional multi-factor authentication.<br>• Sessions should expire after a period of inactivity and immediately on sign out.<br>• Repeated failed login attempts should be rate limited. | SR1, SR2 | H1.2 |
| User Information | Profile or health data is visible to other users (High) | • Sensitive health information such as weight, height, and fitness goals is exposed to other users without consent.<br>• Users may feel embarrassed or judged by friends who can see their personal data.<br>• Flexin' may be in violation of privacy regulations governing health data. | • Database access rules are too broad, allowing users to read other users' profile records.<br>• Social screens fetch a user's entire profile instead of only the fields needed for display.<br>• Default privacy settings expose data the user never chose to share. | • Profiles should be private by default, with users opting in to share specific information.<br>• Database access rules should follow least privilege, granting each component only the data it needs.<br>• Users should be able to set the visibility of each profile field individually. | SR3 | H2.1 |

| Component | Failure Mode (Risk) | Effects of Failure(s) | Causes of Failure(s) | Recommended Action(s) | SR | Ref. |
|---|---|---|---|---|---|---|
| Fitness Device Integrations | Workout sync fails silently (Medium) | • Workouts the user completed do not appear in Flexin', so they receive no credit or currency for them.<br>• The user's streak breaks through no fault of their own, causing frustration and a sense of failure.<br>• The user may stop trusting that their effort will be recognized and disengage from the app. | • The third-party health API experiences a temporary outage or returns an error.<br>• The user revokes the permissions Flexin' needs, or the permissions expire.<br>• The device's operating system terminates Flexin's background sync to save battery. | • The app should display the last successful sync time and any sync errors.<br>• Failed syncs should be retried automatically, and users should be able to trigger a manual re-sync.<br>• Streaks should have a grace period so that a delayed sync does not immediately break them. | SR4, SR5, SR8 | H3.1 |
| Exercise Data | Repeatedly missed goals are presented to the user as failure (High) | • The user experiences shame, guilt, and lowered self-esteem from repeatedly being told they have failed.<br>• The user may come to associate exercise with failure and abandon it entirely, defeating the purpose of Flexin'.<br>• Users already struggling with body image or motivation may have those struggles made worse. | • Goals are set unrealistically high for the user's current fitness level.<br>• The app uses punitive messaging and visuals, such as broken streaks and "missed" markers.<br>• The system does not account for illness, injury, or planned rest. | • All messaging around missed goals should be supportive and focus on progress rather than failure.<br>• Users should be able to use streak freezes and schedule rest days.<br>• After repeated misses, the app should suggest lowering the goal to a more achievable level.<br>• Missed goals should never be shown to other users. | SR6, SR7, SR8, SR9 | H4.1 |
| Exercise Data | Automatic goal progression pushes users to overexert themselves (High) | • The user pushes beyond their physical limits to meet goals, increasing the risk of injury.<br>• Constantly rising targets lead to physical and mental burnout.<br>• An injured or burnt-out user may be unable to exercise for an extended period, losing the habit Flexin' aims to build. | • Goals increase automatically after each success with no upper bound.<br>• The progression model does not build in rest or recovery days.<br>• The system does not consider the user's fitness level or recent activity when raising goals. | • The rate at which goals can increase should be limited.<br>• Rest days should be built into goal progression.<br>• Users should always be able to manually lower or override their goals. | SR6 | H4.2 |
| Currency Attribution | Reward structure incentivizes unhealthy behaviour (High) | • Users continue training while sick or injured to avoid losing their streak, worsening their condition.<br>• Users exercise excessively to earn more currency, increasing the risk of injury.<br>• Exercise becomes driven by the reward rather than health, undermining the habit Flexin' aims to build. | • Rewards scale with activity volume without any limit.<br>• Streaks reset completely after a single missed day, making them feel too valuable to lose.<br>• The reward system does not distinguish between healthy and excessive activity. | • A daily cap should limit how much currency can be earned.<br>• Rewards should diminish for activity beyond a healthy daily volume.<br>• Users should be able to freeze their streaks when sick or injured. | SR8 | H5.1 |

| Component | Failure Mode (Risk) | Effects of Failure(s) | Causes of Failure(s) | Recommended Action(s) | SR | Ref. |
|---|---|---|---|---|---|---|
| Market Simulation | Simulated market returns are unrealistic (High) | • Users develop a false understanding of risk, believing markets grow steadily without significant losses.<br>• Users overestimate how quickly compounding grows their money.<br>• The core educational purpose of Flexin' is undermined, and users may carry these misconceptions into real-world investing. | • The price model is too smooth and does not produce realistic volatility.<br>• Simulated returns are tuned higher than historical market averages to feel more rewarding.<br>• The model never produces significant downturns or crashes. | • The simulation model should be calibrated against historical market data.<br>• The simulation should include realistic volatility and periodic drawdowns.<br>• The limitations of the model should be documented and explained within the app. | SR10, SR11 | H6.1 |
| Investment Platform | Users treat results in Flexin' as real-world truth and apply strategies to real money (High) | • Users invest real money expecting the returns they saw in Flexin', and may suffer real financial loss when markets behave differently.<br>• Users may overconcentrate in a single stock that performed well in the simulation.<br>• Users may overlook real-world costs such as fees and taxes that the simulation does not model. | • The simulation looks and feels like a real trading platform, giving it unwarranted authority.<br>• There is no clear disclaimer that results are simulated.<br>• The app frames gains as the result of skill rather than luck and time in the market. | • A persistent disclaimer should state that Flexin' is a simulation and not financial advice.<br>• The app should explain what the simulation does not model, such as fees, taxes, and real-world volatility.<br>• Educational content should frame results as the product of time and chance rather than skill. | SR11 | H7.1 |
| Investment Platform | Trade only partially completes (High) | • The user's currency is deducted without the purchased shares being added, or shares are added without payment.<br>• The user's balance and portfolio no longer reflect their actual activity, misrepresenting their progress.<br>• Users lose trust in the accuracy of the platform. | • The trade is written to the database as separate, non-atomic operations.<br>• A network failure or app crash interrupts the transaction partway through.<br>• The price changes between when the user views it and when the trade executes. | • Each trade should be executed as a single atomic database transaction.<br>• The price should be captured at the moment the user confirms the trade.<br>• Incomplete transactions should be detected and rolled back automatically. | SR12 | H7.2 |
| Social Management | Leaderboards and group comparison cause negative psychological effects (High) | • Users who rank poorly experience shame and anxiety, especially when losing to friends.<br>• Users feel compelled to compete constantly, leading to unhealthy exercise or compulsive app usage.<br>• Lower-ranked users may leave their groups or the platform entirely. | • Leaderboards rank users by absolute performance, favouring those who are already fit.<br>• Missed goals and low rankings are visible to the entire group.<br>• Users have no option to step back from competition. | • Users should be able to opt out of leaderboards.<br>• Rankings should be based on progress relative to each user's own goals.<br>• Missed goals should remain private to the user. | SR3, SR9 | H8.1 |

| Component | Failure Mode (Risk) | Effects of Failure(s) | Causes of Failure(s) | Recommended Action(s) | SR | Ref. |
|---|---|---|---|---|---|---|
| Database | Security rules expose all users' data (High) | • The health and profile data of every Flexin' user can be read or modified by unauthorized parties.<br>• Flexin' may face legal consequences under privacy regulations governing health data.<br>• Users lose trust in the platform and leave. | • Database access rules are misconfigured or overly permissive.<br>• Rules are changed during development without being re-tested.<br>• Test or debug configurations are accidentally deployed to production. | • Database access rules should deny all access by default and grant only what is required.<br>• Automated security rule tests should run in CI before every release.<br>• Production and development configurations should be kept separate. | SR3 | H9.1 |
| Database | Data is lost or corrupted (High) | • Users lose their workout history, streaks, currency balances, and portfolios.<br>• Corrupted data may cause incorrect rewards or broken app behaviour.<br>• Users who lose long-term progress may feel discouraged and abandon the platform. | • A multi-step write fails partway through, leaving data in an inconsistent state.<br>• A schema migration contains an error that corrupts or drops existing records.<br>• Data is deleted accidentally by a bug or an administrative mistake. | • All multi-step writes should be wrapped in transactions.<br>• Migrations should be tested against a copy of real data before deployment.<br>• Regular backups should be taken, and restoring from them should be tested. | SR12 | H9.2 |
| User Interface & Experience | Engagement design becomes addictive or guilt-inducing (Medium) | • Users develop compulsive habits of checking the app, similar to social media addiction.<br>• Guilt-based reminders increase stress and anxiety around exercise.<br>• Users may disable notifications entirely or uninstall the app. | • The app sends excessive notifications to maximize engagement.<br>• Reminders use guilt-based language that pressures the user.<br>• Streak and reward mechanics are designed to encourage constant checking. | • Users should control how often they receive notifications and be able to set quiet hours.<br>• Reminders should use encouraging language and never shame the user.<br>• Engagement features should be reviewed against the hazards in this analysis before release. | SR7 | H10.1 |
| Fitness Device Integrations, Exercise Data, Currency Attribution | A late sync attributes a workout to the wrong day (Medium) | • The user's streak breaks even though they exercised on the correct day.<br>• The user loses the currency they earned for that day.<br>• The user feels they have failed despite meeting their goal, contributing to H4.1. | • Workouts are dated by the time they sync rather than the time they occurred.<br>• The device syncs after midnight, placing the workout in the following day.<br>• Time zone differences between the device and server are handled incorrectly. | • Workouts should be dated by when they occurred, using the timestamp from the source device.<br>• Streaks and rewards should be recalculated when late data arrives.<br>• All timestamps should be stored in UTC and converted to the user's local time for display. | SR5, SR8 | H11.1 |
| Account Creation & Authentication, Fitness Device Integrations | Health data from one person is linked to another user's account (Medium) | • One person's health data is visible to another user, breaching their privacy.<br>• The receiving user gets rewards and progress for workouts they did not do.<br>• The person who actually exercised receives no credit for their workouts. | • Multiple people share a single device or health platform account.<br>• Linked devices are not unlinked when the user signs out.<br>• The app does not show which account a device is currently linked to. | • Linked devices should be unlinked automatically when the user signs out.<br>• The app should clearly display which Flexin' account a device is linked to.<br>• Users should be able to view and remove linked devices from their settings. | SR2 | H11.2 |

# 6 Safety and Security Requirements

The requirements below address the hazards identified in Table 2. Each requirement lists the hazards it mitigates, and its priority is the highest risk level among those hazards. These requirements will also be added to the SRS.

SR1: *The system shall enforce a minimum password strength, offer optional multi-factor authentication through the user's email, and temporarily lock sign-in after repeated failed attempts while allowing the user to reset their password through email.*
**Rationale:** Weak passwords and unlimited login attempts allow attackers to access users' private data, while legitimate users must still be able to regain access to their accounts.
**Fit Criterion:** Passwords that do not meet the strength policy must be rejected, and sign-in must be temporarily locked after repeated consecutive failed attempts.
**Associated Hazards:** H1.1, H1.2
**Priority:** High

SR2: *The system shall end a user's session after a period of inactivity or when the user signs out, and shall unlink the user's fitness devices on sign out.*
**Rationale:** Sessions and device links left open on shared or lost devices allow other people to access an account or send their health data to it.
**Fit Criterion:** Sessions must expire after a period of inactivity, and no session or device link may remain active after the user signs out.
**Associated Hazards:** H1.2, H11.2
**Priority:** High

SR3: *The system shall keep all profile and health data private by default, sharing a field with other users only after the user opts in, and shall enforce this through default-deny database access rules.*
**Rationale:** Sensitive information such as weight, goals, and missed workouts must never be exposed to other users or unauthorized parties without consent.
**Fit Criterion:** All sharing settings must be disabled for new accounts, and automated security rule tests run in CI must confirm that no user can read another user's unshared data before each release.
**Associated Hazards:** H2.1, H8.1, H9.1
**Priority:** High

SR4: *The system shall display a clear error message when account creation, sign-in, or a workout sync fails, and shall automatically retry failed workout syncs.*
**Rationale:** Users should know when and why something has failed, rather than being locked out or discovering missing workouts through a broken streak.
**Fit Criterion:** Every failed account creation, sign-in, or sync must display an error message stating the cause, and failed syncs must be retried before being reported to the user.
**Associated Hazards:** H1.1, H3.1
**Priority:** Medium

SR5: *The system shall date each workout by the time it occurred on the source device, and shall recalculate streaks and rewards when workout data arrives late.*
**Rationale:** Users who exercised on time should not lose their streak or currency because

their device synced late or across a day boundary.
**Fit Criterion:** Workouts synced after midnight or from a different time zone must be recorded on the day they occurred, and affected streaks and rewards must be updated once the late data arrives.
**Associated Hazards:** H3.1, H11.1
**Priority:** Medium

SR6: *The system shall limit how quickly automatic goals increase, build rest days into goal progression, suggest lowering a goal that is repeatedly missed, and allow users to manually lower or override any goal.*
**Rationale:** Goals that rise too quickly push users to overexert themselves, while goals that are repeatedly missed lead to shame and abandonment of exercise.
**Fit Criterion:** Automatic goals must increase gradually and include regular rest days, and a lower goal must be suggested after a goal is missed several times in a row.
**Associated Hazards:** H4.1, H4.2
**Priority:** High

SR7: *The system shall use supportive, non-punitive language in all goal-related messages and reminders, and shall allow users to control notification frequency and set quiet hours.*
**Rationale:** Guilt-based messaging and excessive notifications harm users' mental health and encourage compulsive app usage.
**Fit Criterion:** All goal-related messages must pass a team review for shaming or guilt-based language, and no notifications may be sent during the user's quiet hours or above their chosen frequency.
**Associated Hazards:** H4.1, H10.1
**Priority:** High

SR8: *The system shall cap the currency a user can earn per day, reduce rewards for activity beyond a healthy daily volume, and allow users to freeze their streak for days missed due to illness, injury, rest, or delayed syncs.*
**Rationale:** Unlimited rewards and unforgiving streaks incentivize users to exercise excessively or while injured, and make users feel they have failed through no fault of their own.
**Fit Criterion:** Users must not be able to earn currency beyond the daily cap, excessive activity must earn a reduced reward rate, and users must be able to freeze their streak.
**Associated Hazards:** H3.1, H4.1, H5.1, H11.1
**Priority:** High

SR9: *The system shall allow users to opt out of group leaderboards, and shall rank users by progress relative to their own goals rather than absolute performance.*
**Rationale:** Comparing users by absolute performance causes shame and anxiety for those who rank poorly.
**Fit Criterion:** Users who opt out must not appear on any leaderboard, and leaderboard rankings must be computed from each user's percentage of their own goal.
**Associated Hazards:** H4.1, H8.1
**Priority:** High

SR10: *The system shall calibrate its market simulation against historical market data, producing volatility and downturns consistent with real markets.*

**Rationale:** Unrealistically smooth or high returns teach users a false understanding of risk and compounding.
**Fit Criterion:** Over a long simulated period, the average annual return and volatility must be consistent with historical broad market index data, and downturns must occur.
**Associated Hazards:** H6.1
**Priority:** High

SR11: *The system shall display a persistent disclaimer stating that Flexin' is a simulation and not financial advice, and shall explain what the simulation does not model, such as fees, taxes, and real-world volatility.*
**Rationale:** Users who treat simulated results as real-world truth may apply them to real investments and suffer real financial loss.
**Fit Criterion:** The disclaimer must be visible on every investment screen, and an explanation of the simulation's limitations must be reachable from every investment screen.
**Associated Hazards:** H6.1, H7.1
**Priority:** High

SR12: *The system shall execute all multi-step database writes, including trades at the price confirmed by the user, as atomic transactions that are rolled back if any step fails, and shall maintain regular backups of all user data.*
**Rationale:** Partial writes and data loss leave users' balances, portfolios, and progress incorrect or permanently lost.
**Fit Criterion:** Tests that simulate a failure at each step of a multi-step write must leave the database unchanged, trades must execute at the confirmed price, and backups must be taken regularly.
**Associated Hazards:** H7.2, H9.2
**Priority:** High

# 7 Roadmap

This roadmap states which safety requirements will be implemented as part of Flexin' within the capstone course timeline and which requirements will be saved for after the course is over.

Requirements that will be implemented as part of the capstone timeline, as they are considered to be high priority, and will be implemented first:

- SR 1
- SR 2
- SR 3
- SR 6
- SR 8
- SR 9
- SR 11

- SR 12

Requirements that will be implemented in the future, as they are considered lower priority, and will only be implemented if time permits:

- SR 4
- SR 5
- SR 7
- SR 10

# 8 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

## 8.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

## 8.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

## 8.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

# 9 Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

# References

[1] D. Boisclair, A. Lusardi, and P.-C. Michaud, "Financial literacy and retirement planning in Canada," *Journal of Pension Economics and Finance*, vol. 16, no. 3, pp. 277–296, 2017, doi: 10.1017/S1474747215000311.